

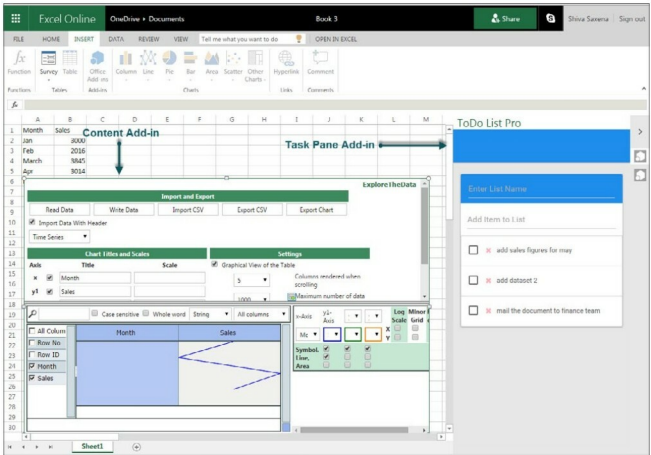


Figure 1: Examples of task pane and content add-ins

to the Office app store or SharePoint app catalogue, and contains metadata about the application like the developer's name, target platform, etc. The Web application can be hosted on any Web server.

Prerequisites for developing add-ins for Office

The recommended IDE for add-ins development is Visual Studio 2015. There is a free version for individual developers and open source contributors (Visual Studio Community Edition) that can be downloaded from <https://www.visualstudio.com/en-us/products/visual-studio-community-vs.aspx>

Download and install the Office Developer Tools extension for Visual Studio from <https://www.visualstudio.com/en-us/features/office-tools-vs.aspx>

Last of all, but most important, you will need Microsoft Office 2013, 2016 or Office 365. If you don't already have one of these, you can download a trial version from the Office website or sign up for the Office 365 Dev program (<http://dev.office.com/devprogram>).

Tip: Alternatively, Napa Development Tools can also be used for Office development. Napa is a free, quick, in-browser tool for developing Office add-ins. Napa tools can be accessed from the website <https://www.napacloudapp.com>

[napacloudapp.com](https://www.napacloudapp.com)

Building a word cloud generator add-in

In this section, we will build a simple word cloud generator task pane add-in for Microsoft Word, which allows users to select a bunch of words from which it generates a word cloud image. It also allows users to insert that word cloud into the document. For the actual word cloud generation, we will take the help of a JavaScript library called *wordcloud2.js* (<https://github.com/timdream/wordcloud2.js>). So, let's get started with the interesting parts.

1. Fire up Visual Studio and go to **File > New > Project**.
2. Although almost all of the development is done in HTML 5, CSS and JavaScript, for some reason, the template for the project is located under the Visual C# section. So navigate through the drop down hierarchy to **Visual C# > Office/SharePoint > Office Add-in**.

3. Select **Office Add-in Project**, name it 'Word Cloud Generator', click OK and in the next screen, select **Task Pane Add-in** as the type.

Now, if you navigate to the Solution Explorer, you will see two projects — 'Word Cloud Generator' and 'Word Cloud Generator Web'. The former contains the *Manifest.xml* file and the latter contains all our HTML 5, CSS and JavaScript files.

4. Navigate to the Word Cloud Generator Project and open